higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

6 December 2024

This marking guideline consists of 7 pages.

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SECTION A: FLOW MEASUREMENT

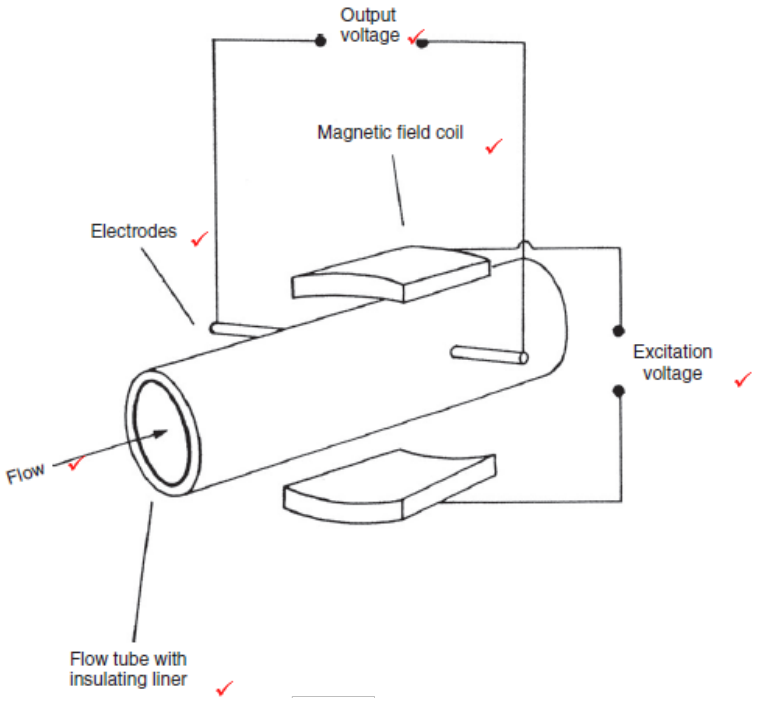
QUESTION 1

1.1 1.1.1
- Glass type variable area flow meter
- Metal tube type variable area flow meter (2)

1.1.2
- Bullet shaped disc
- Mushroom- shaped disc
- Plugged – shaped disc (3)

1.2 1.2.1 Faraday's law of electromagnetic induction states that the voltage induced in any conductor as it moves across a magnetic field✓ is proportional to the velocity of that conductor,✓ magnetic field strength and the length of the conductor.✓ (3)

1.2.2



1 mark for sketch (7)

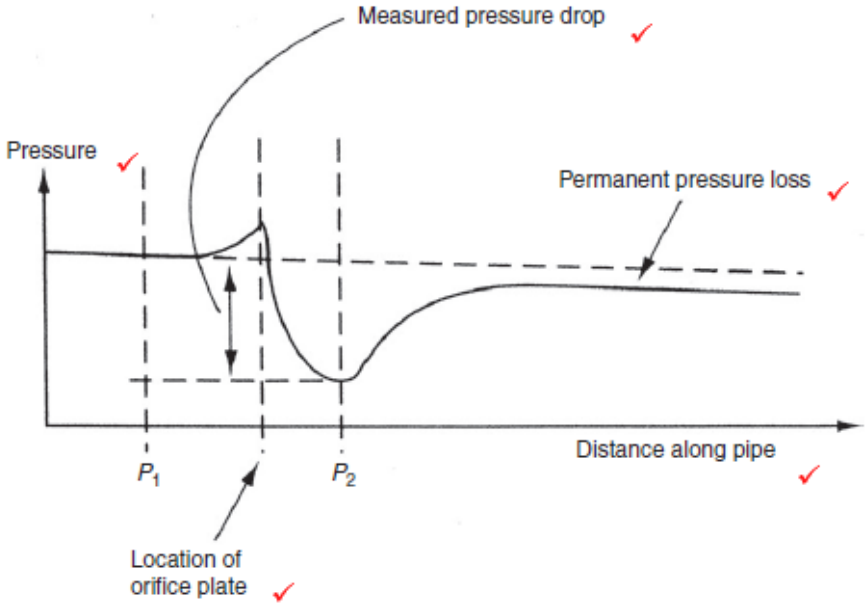
1.2.3 By Faraday's law of electromagnetic induction, the voltage, E, induced across a length is given by formula E = BLv,✓✓ where ;the flowing fluid moving at velocity = v,✓ in a magnetic field of flux density = B✓ and L is the distance between the electrodes.✓ (5)
[20]

QUESTION 2

2.1 The function of an integrator is to measure the total volume of flow in a certain time.✓✓ (2)

2.2 Electromagnetic flowmeters are designed to measure liquid flow,✓ thus cannot measure compressible fluids✓ like gases. (2)

2.3



(5)

2.4 2.4.1 The upstream pressure tapping is taken one pipe diameter, D, upstream from the orifice plate✓ and the downstream pressure tapping is taken at one half pipe diameter D/2, downstream from the orifice plate.✓

2.4.2 The upstream tapping is the one pipe diameter from the upstream face of the orifice plate✓ and the downstream tapping is determined from a curve relating the required dimension to the ratio of the orifice to pipe diameter.✓

2.4.3 Are taken via holes cut obliquely through a flange or boss on the pipe.✓✓

(3 × 2) (6)

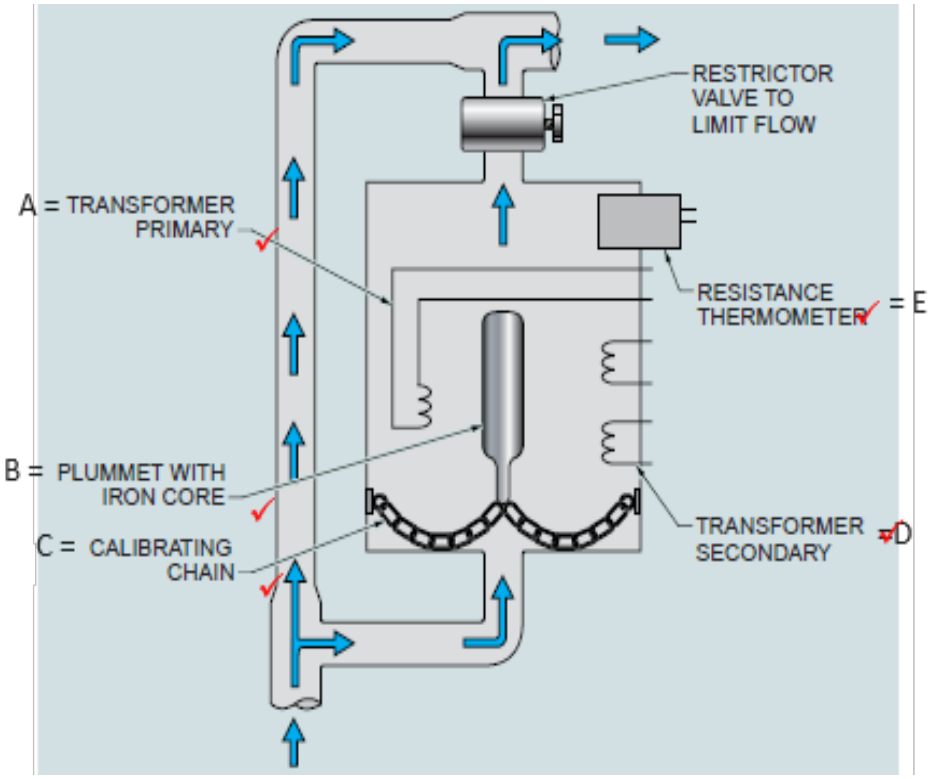
[15]

TOTAL SECTION A: 35

SECTION B: DENSITY, HUMIDITY & VISCOSITY MEASUREMENT

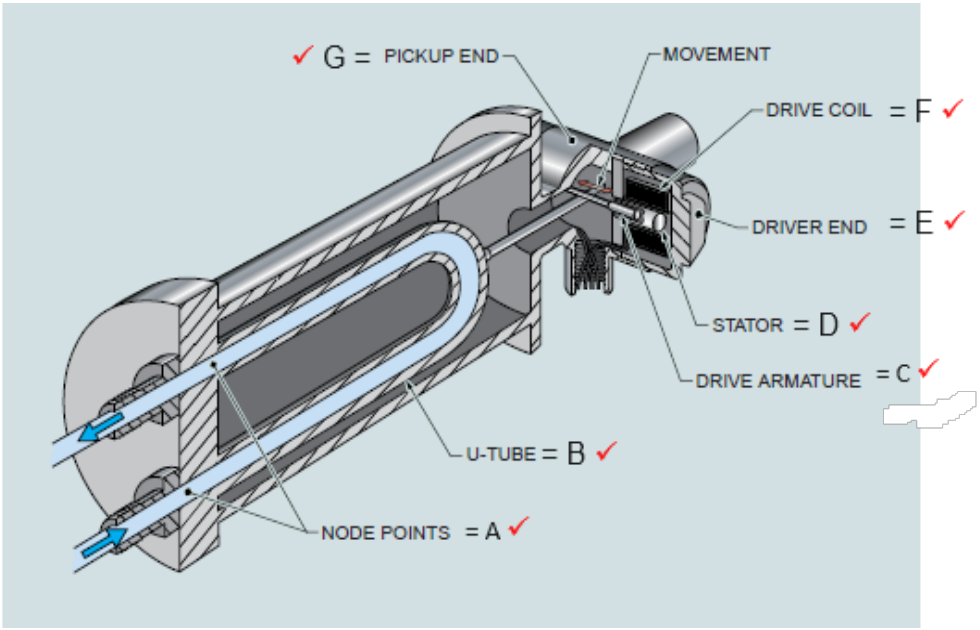
QUESTION 3

3.1



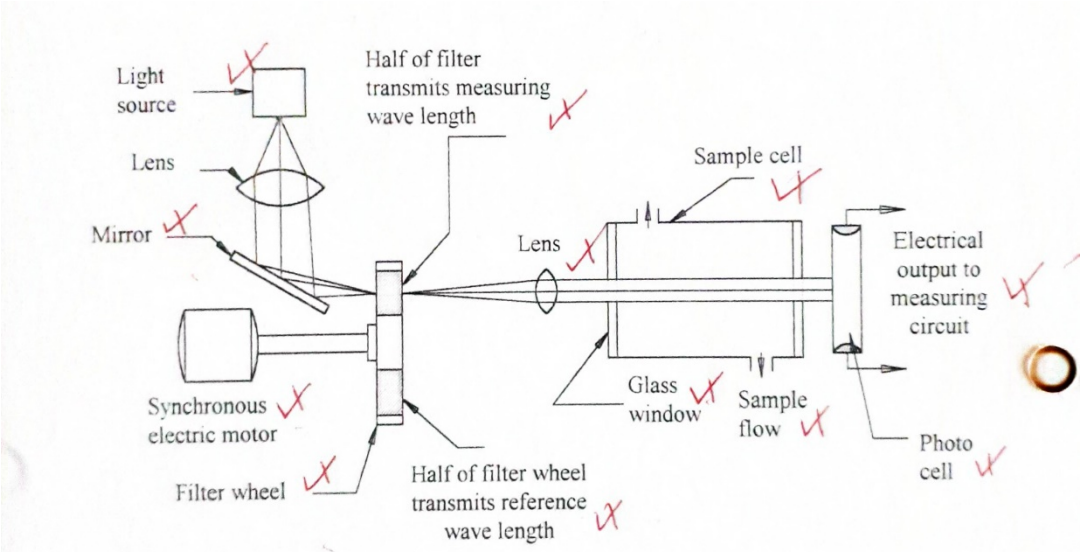
(5 × 1) (5)

3.2



(7 × 1) (7)

3.3



(6)

3.4 3.4.1 Electrolytic hygrometer (1)

3.4.2
- The principle of measurement involves the electrolysis of water into oxygen and hydrogen.✓
- Since two electrons are required for the electrolysis of each water molecule,✓ the electrolysis current is a measure of water present in the sample.✓
- The electrolysis current is a function of water concentration in the sample.✓
- In order to obtain output in the form of moisture content, the sample flowrate must be a constant value known.✓ (5)

[24]

QUESTION 4: pH AND CONDUCTIVITY MEASUREMENT

4.1 4.1.1
- Those which directly amplify the electrode EMF ✓ so that the output can operate a standard indicator,✓ recorder and controller.
- Those which compare the electrode EMF with a known standard emf.✓✓ (2 × 2) (4)

4.1.2
- The instrument to be substantially free from errors caused by supply voltage or frequency variation.
- The zero reading to be stable over long periods.
- The current taken from the electrode system to be extremely small to avoid polarization effects.
- The design to include compensation for variation in characteristics of components or be such that the output is independent of such variation.
- Temperature compensation to be included, either manually or automatically.
- Facilities to be included for easy checking of calibration.

(Any 5 × 1) (5)

4.3 4.3.1 The glass electrode system encounters electrical leakage✓ due to moisture on the electrode bodies.✓

4.3.2 The use of silicone coating on the glass electrode✓ prevent leakages ,thus improving measurements.✓

(2 × 2) (4)

4.4
- Washing with water and wiping with cotton wool soaked in diluted hydrochloric acid will suffice.
- In severe cases the electrodes may require soaking in solvents.
- Inorganic solvents are preferable for removing greasy and oily deposits.
- A site cleaning device which provides a jet of solvents to be periodically directed at the electrode may be usedfor frequent cleaning.
- An ultrasonic transducer attached to a thin plate may be employed.
- Organic solvents have a dehydrating effect on the membrane and therefore only brief immersion in a solvent followed by soaking in hydrochloric acid should be employed. (6)

[19]

QUESTION 5: AUTOMATIC CONTROL

5.1 5.1.1 Proportional control (1)

5.1.2 A potential appears at point A, forms an input signal to the amplifier✓ and thus a feedback voltage is generated to C_{FB}✓ to counter the potential at A and the potential at A returns to zero.✓ (3)

5.1.3 The proportional band is adjusted✓ and this will adjust the amount of feedback voltage to C_{FB}.✓ (2)

5.2
- On-off control action
- Two position action
- Two position differential-gap control action
- Multi-position control action (Any 3 × 1) (3)

5.3
- Pressure disturbance values are small.
- There is no steady flow of air in the line.
- There is a uniform distribution of fluid resistance, capacitance and inductance parameters throughout the line.
- The load is lumped.
- The small disturbances that do occur do not vary the mean pressure level in the line. (5)

5.4 5.4.1 A mode of control in which the final control element is quickly moved to only one of the few positions,✓ when the measured variable reached a predetermine magnitude.✓

5.4.2 Increases the output pressure✓ for an increase in the measured variable.✓

5.4.3 A retardation with or without delay in the response of the measured variable✓ to a change in one of the conditions affecting the process.✓

5.4.4 A two position control in which the final control element remains in its last position✓ until the measured variable has moved slightly beyond the control point.✓

(4 × 2) (8)
[22]

TOTAL SECTION B: 65
GRAND TOTAL: 100

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